

Subject: Software Engineering
Assignment 10: Hands-on with Project Management Tools

Introduction

Project management tools are essential in modern software development for organizing tasks, tracking progress, and ensuring timely delivery. In this assignment, tools like **Trello** and **Jira** are used to simulate project management activities for the *Traffic Prediction System (Pune/PMC)*.

The Traffic Prediction System is a full-stack web application designed to predict and visualize real-time traffic congestion in Pune using live sensor data from the Pune Municipal Corporation (PMC). It features an interactive map interface built with React + Vite, a Node.js/Express backend, MongoDB for data storage, Leaflet.js and Google Maps API for mapping, and a dual-scoring traffic prediction engine.

Key Concepts of Project Management Tools

- **Project Tracking:** Monitoring the progress of tasks and ensuring they are completed within deadlines.
- **Task Assignment:** Distributing work among team members based on their roles and responsibilities.
- **Progress Monitoring:** Evaluating how much work has been completed and how much is remaining.
- **Collaboration:** Allowing team members to communicate and work together efficiently.
- **Agile Methodology:** An approach that divides work into smaller iterations called sprints.

Implementation Using Trello

Trello is a Kanban-based tool that uses boards, lists, and cards to organize tasks visually. A board was created for the Traffic Prediction System with the following lists: *Backlog*, *To Do*, *In Progress*, *Blocked*, *Done*.

Task Creation (Trello)

Task ID	Description
TP-01	Dual-scoring prediction formula implementation
TP-02	Flask ML microservice setup
TP-06	Google Maps integration with traffic layer
TP-07	ArcGIS multi-layer integration
TP-13	Node.js REST API development
TP-16	React dashboard frontend development

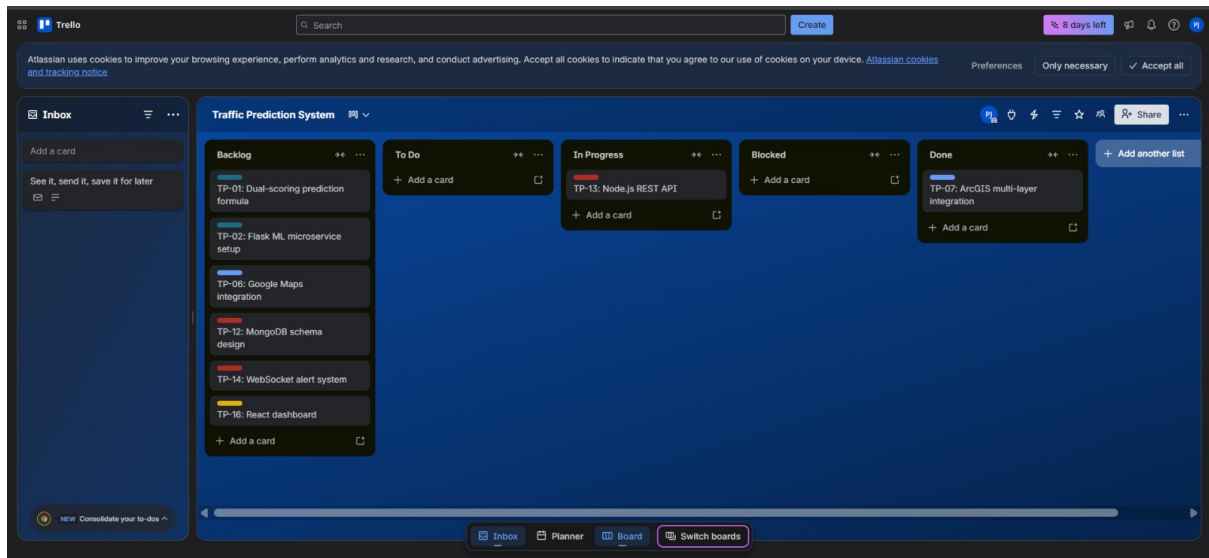


Figure 1: Trello Board — Traffic Prediction System Workflow

Implementation Using Jira

Jira is an advanced tool used for Agile development, supporting Scrum and Kanban workflows.

Backlog and Issue Management

Tasks were added to the backlog as Stories and Bugs, categorized into Epics such as *Map & GIS Integration* and *Backend & ML Engine*.

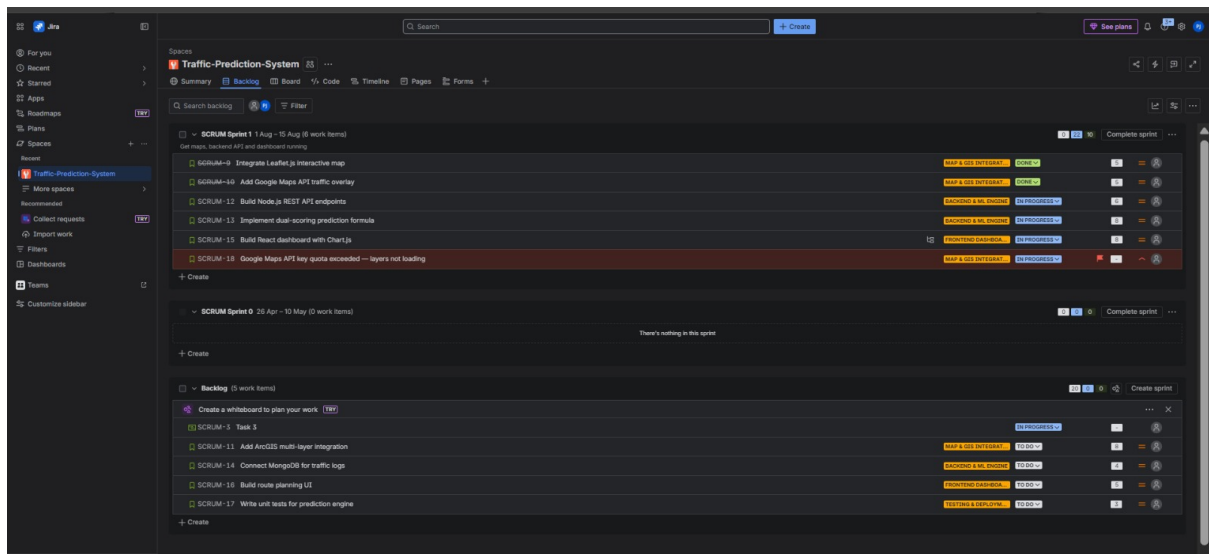


Figure 2: Jira Backlog View — Sprint 1 Planning

Issue Details and Subtasks

Each major task was broken down into subtasks for granular tracking and assigned to team members.

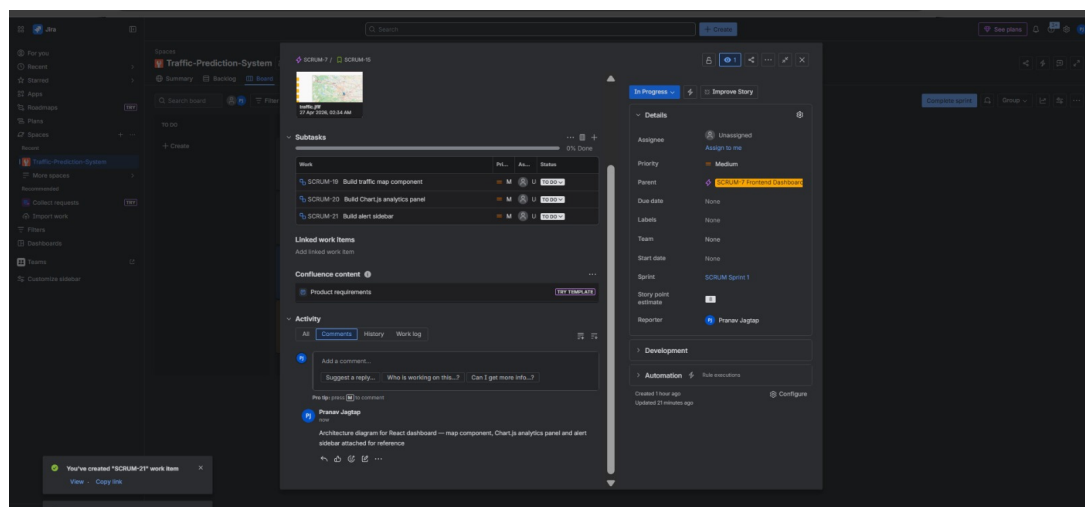


Figure 3: Jira Issue Details and Subtask Management

Bug and Blocker Tracking

Critical issues, such as API quota limitations, were flagged as high-priority blockers to ensure immediate attention.

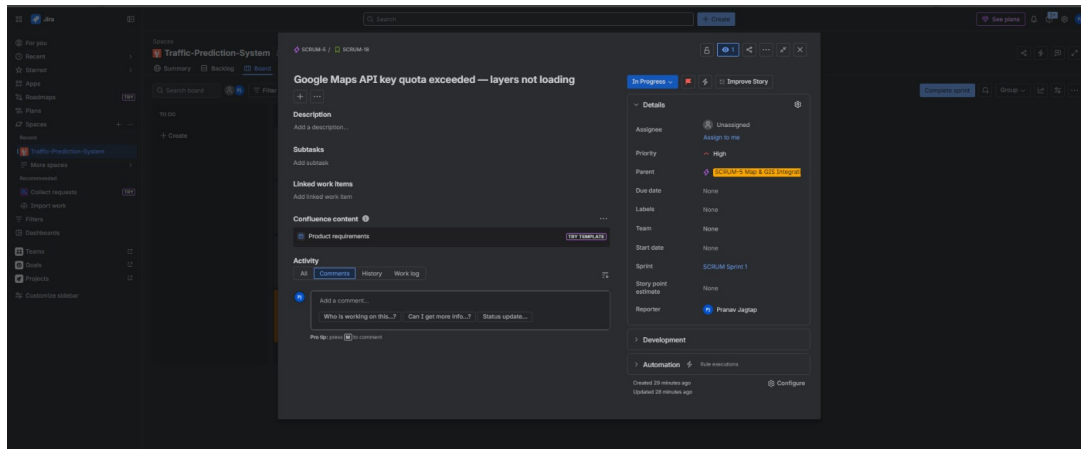


Figure 4: Jira Bug Tracking — Handling API Blocker

Active Sprint Board

The sprint board was used to monitor daily progress, moving items from *To Do* to *Done*.

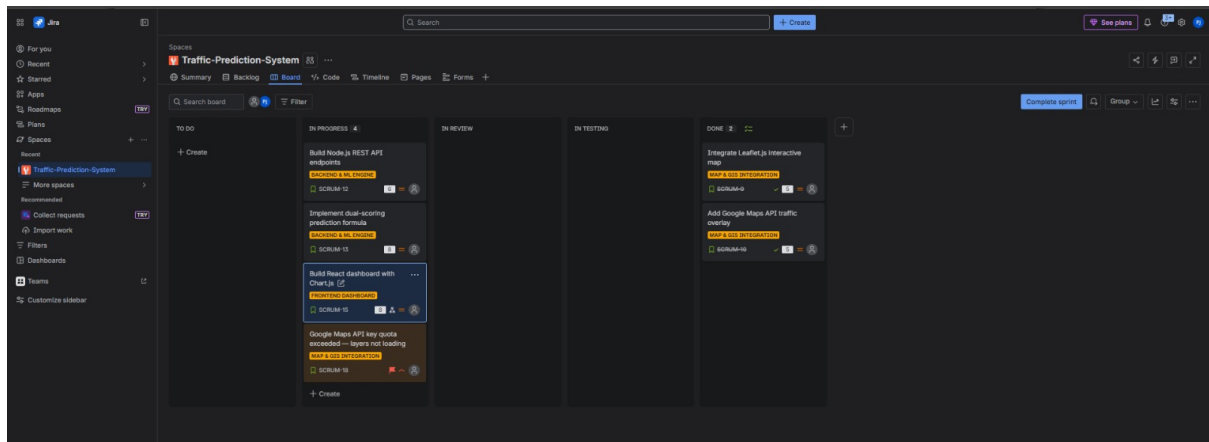


Figure 5: Jira Active Sprint Board (Kanban/Scrum view)

Task Distribution (Group 16)

Task	Assigned To	Status
Leaflet.js map integration	Team Member	Done
Google Maps API overlay	Team Member	Done
Node.js REST API	Team Member	In Progress
Dual-scoring formula	Team Member	In Progress
React dashboard	Team Member	In Progress
ArcGIS integration	Team Member	To Do

Comparison: Trello vs Jira

Feature	Trello	Jira
Ease of Use	Very Easy	Moderate
Interface	Visual (Kanban)	Agile-based (Scrum/Kanban)
Best For	Small projects	Large/Complex projects
Features	Basic	Advanced (Reporting, Epics)
Tracking	Simple movement	Detailed status & burndown

Conclusion

The implementation of Trello and Jira provided valuable experience in project management. Trello is ideal for its simplicity and visual layout, while Jira offers the powerful features needed for professional Agile software development. Both tools significantly improve coordination and productivity in software engineering projects.